Linear Algebra II — TF2

Section 5.4

13. Let T be a linear operator on a vector space V, let v be a nonzero vector in V, and let W be the T-cyclic subspace of V generated by v. For any $w \in V$, prove that $w \in W$ if and only if there exists a polynomial $g(t)$ such that $w = g(T)(v)$.

21. Let T be a linear operator on a two-dimensional vector space V. Prove that either V is a T-cyclic subspace of itself or $T = cI$ for some scalar c.

Section 6.1

17. Let T be a linear operator on an inner product space V, and suppose that $||T(x)|| = ||x||$ for all x. Prove that T is one-to-one.

18. Let V be a vector space over F, where $F = R$ or $F = C$, and let W be an inner product space over F with inner product $\langle \cdot, \cdot \rangle$. If $T : V \longrightarrow W$ is linear, prove that $\langle x, y \rangle' = \langle T(x), T(y) \rangle$ defines an inner product on V if and only if T is one-to-one.

Section 6.2

7. Let β be a basis for a subspace W of an inner product space V, and let $z \in V$. Prove that $z \in W^{\perp}$ if and only if $\langle z, v \rangle = 0$ for every $v \in \beta$.

11. Let A be an $n \times n$ matrix with complex entries. Prove that $AA^* = I$ if and only if the rows of A form an orthonormal basis for C^n.

15. Let V be a finite-dimensional inner product space over F.

    (a) *Parseval's Identity.* Let $\{v_1, v_2, ..., v_n\}$ be an orthonormal basis for V. For any $x, y \in V$ prove that

$$\langle x, y \rangle = \sum_{i=1}^{n} \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$